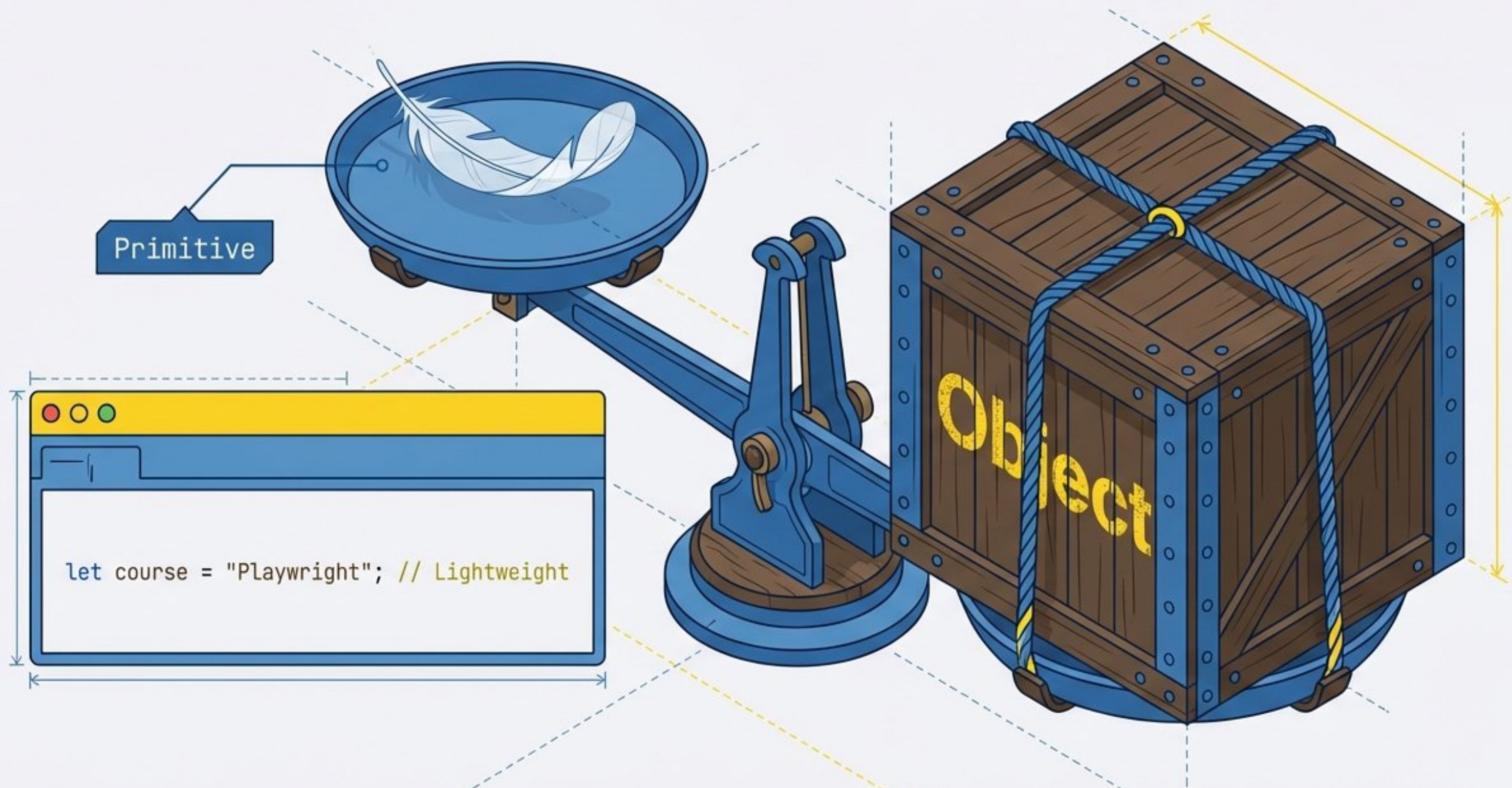


Playwright Training

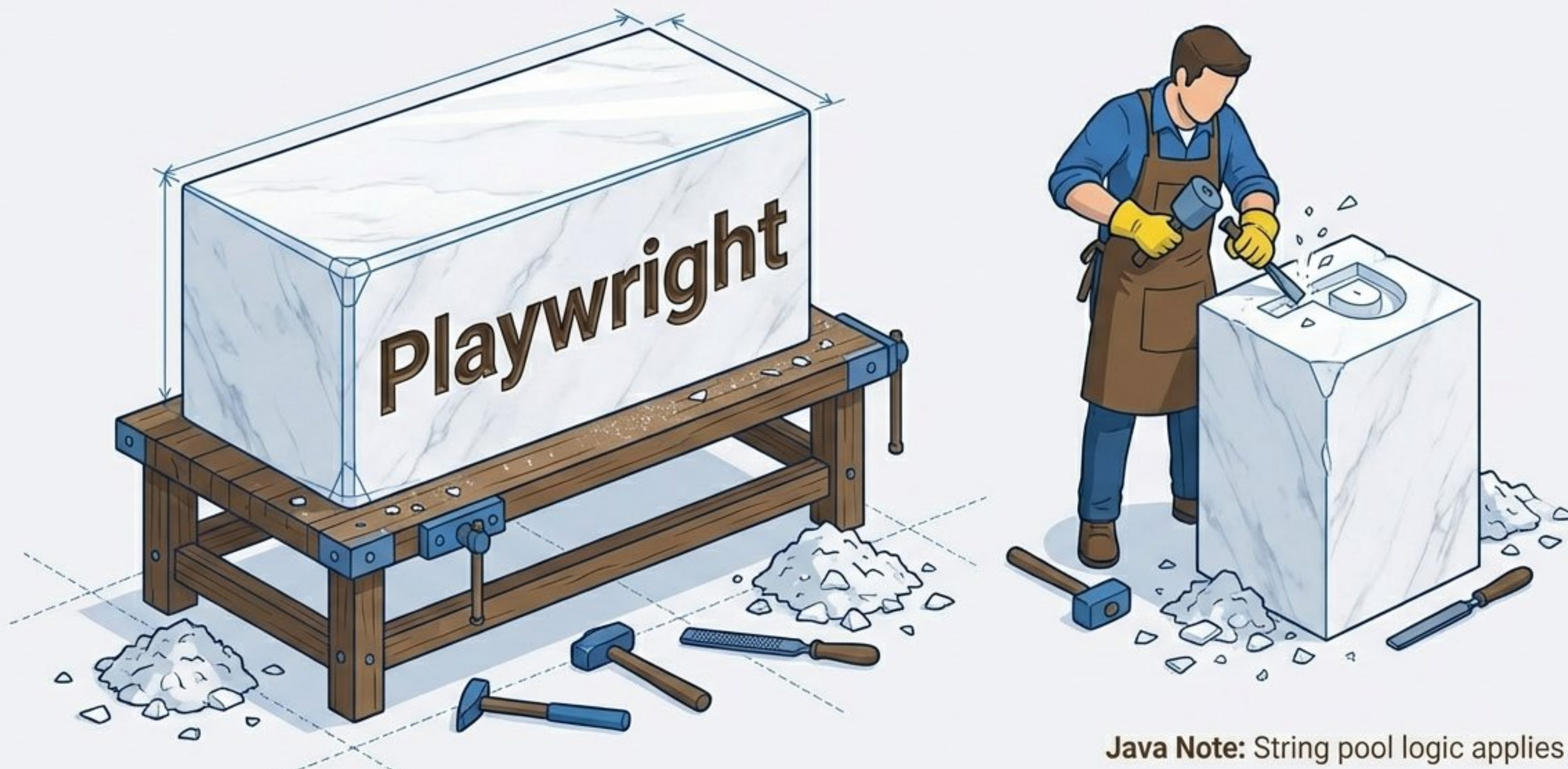
String

Primitives > Objects



Immutable by Design

Cannot change in-place. Must create a copy.



Java Note: String pool logic applies here too.

JavaScript String – Common Helper Methods

✓ Length & Case Conversion

→ **length** → Returns total number of characters

`"Play".length` → 4

`toUpperCase()` / `toLowerCase()`

Aa

Change Case

`trim()`



Remove Whitespace

`includes()`



Contains Check

`startsWith()` / `endsWith()`



Check Start / End

`indexOf()`



Find Position

`split()`



Split String

`replace()` / `replaceAll()`



Find & Replace

`charAt()`

A

Get Character

`concat()`



Join Strings

✓ Extracting Text

→ **slice**(start, end) → Extracts part (supports index)

`"Playwright".slice(0,4)` → "Play"

→ **substring**(start, end) → Extracts part (no negative support)

`"Playwright".substring(0,4)` → "Play"

✓ **Difference:** slice supports negative values

✓ Transforming Strings

→ **split**() → Converts string to array

`"a,b,c".split(",")` → ["a","b","c"]

→ **replace**() → Replaces first match

`"hi hi".replace('hi','hello')` → "hello hi"

i Character Access & Combining

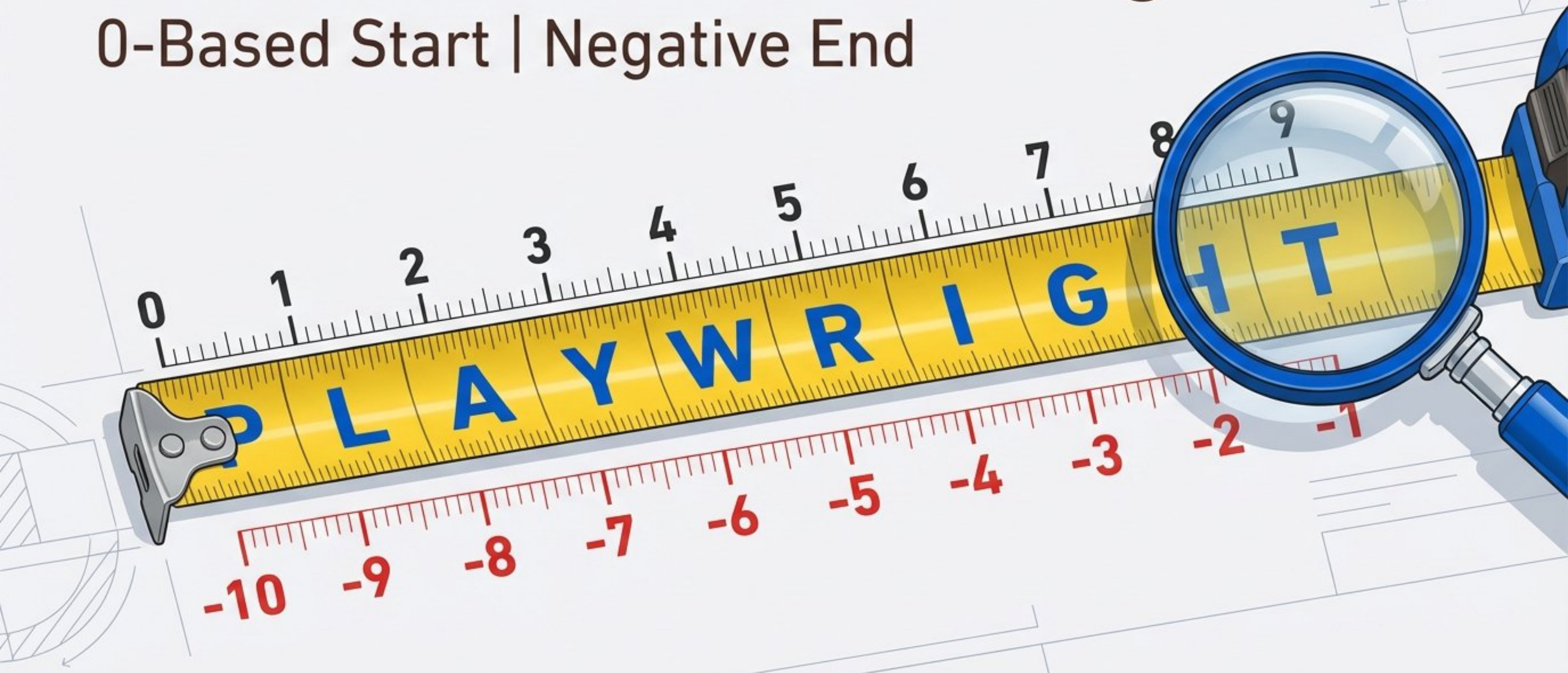
→ **charAt**(index) → Returns character at position

`"JS".charAt(0)` → "J"

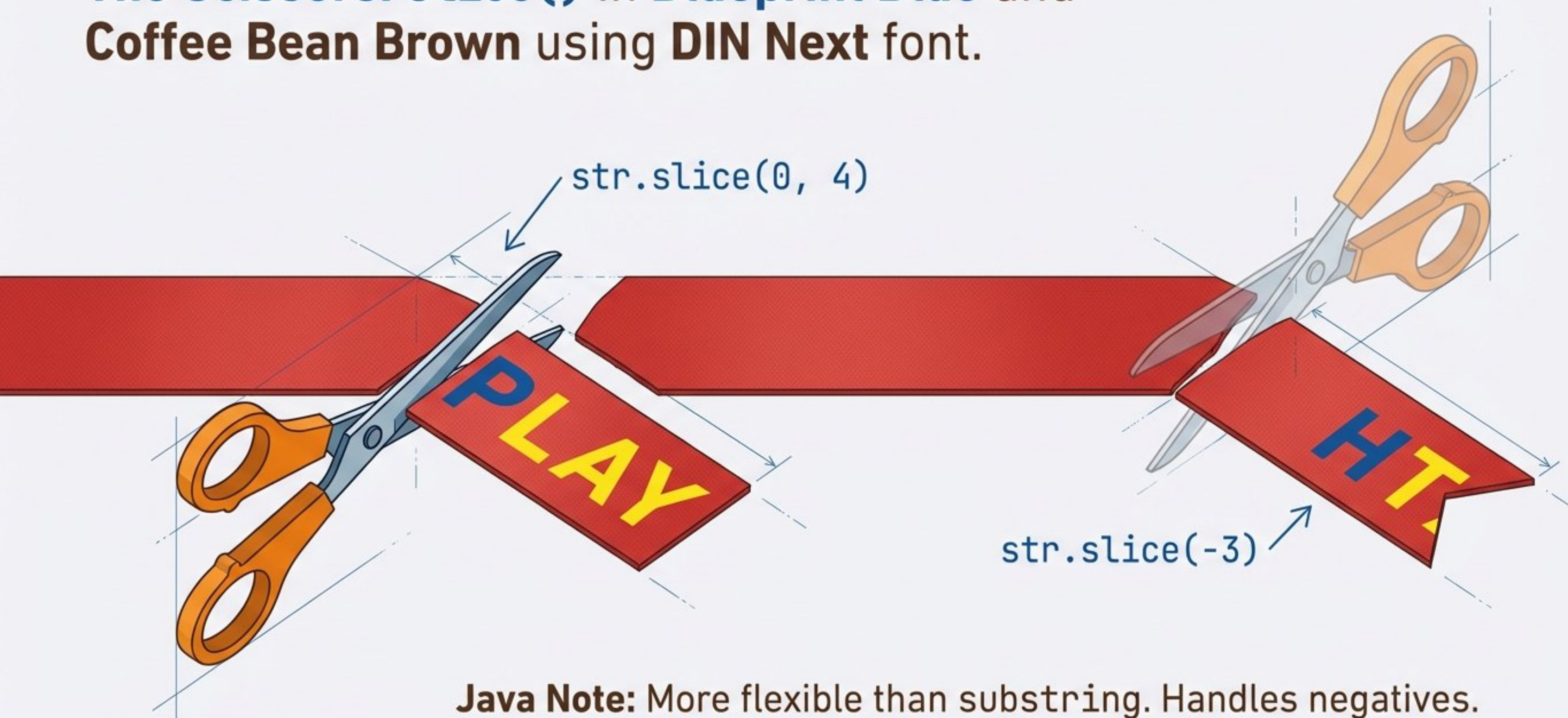
→ **concat**() → Combines strings

The Ruler: Dual Indexing

0-Based Start | Negative End

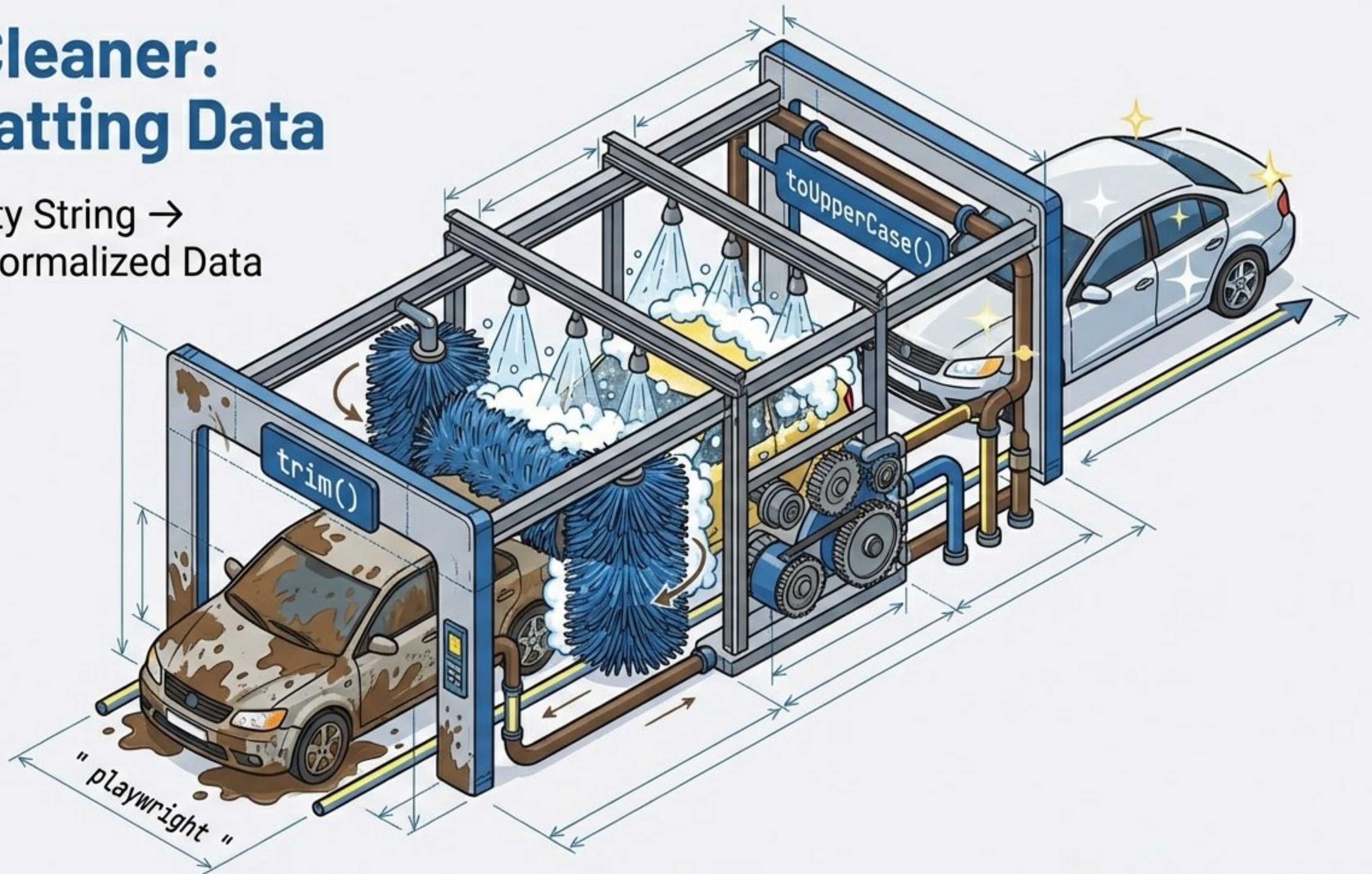


The Scissors: `slice()` in **Blueprint Blue** and **Coffee Bean Brown** using **DIN Next** font.

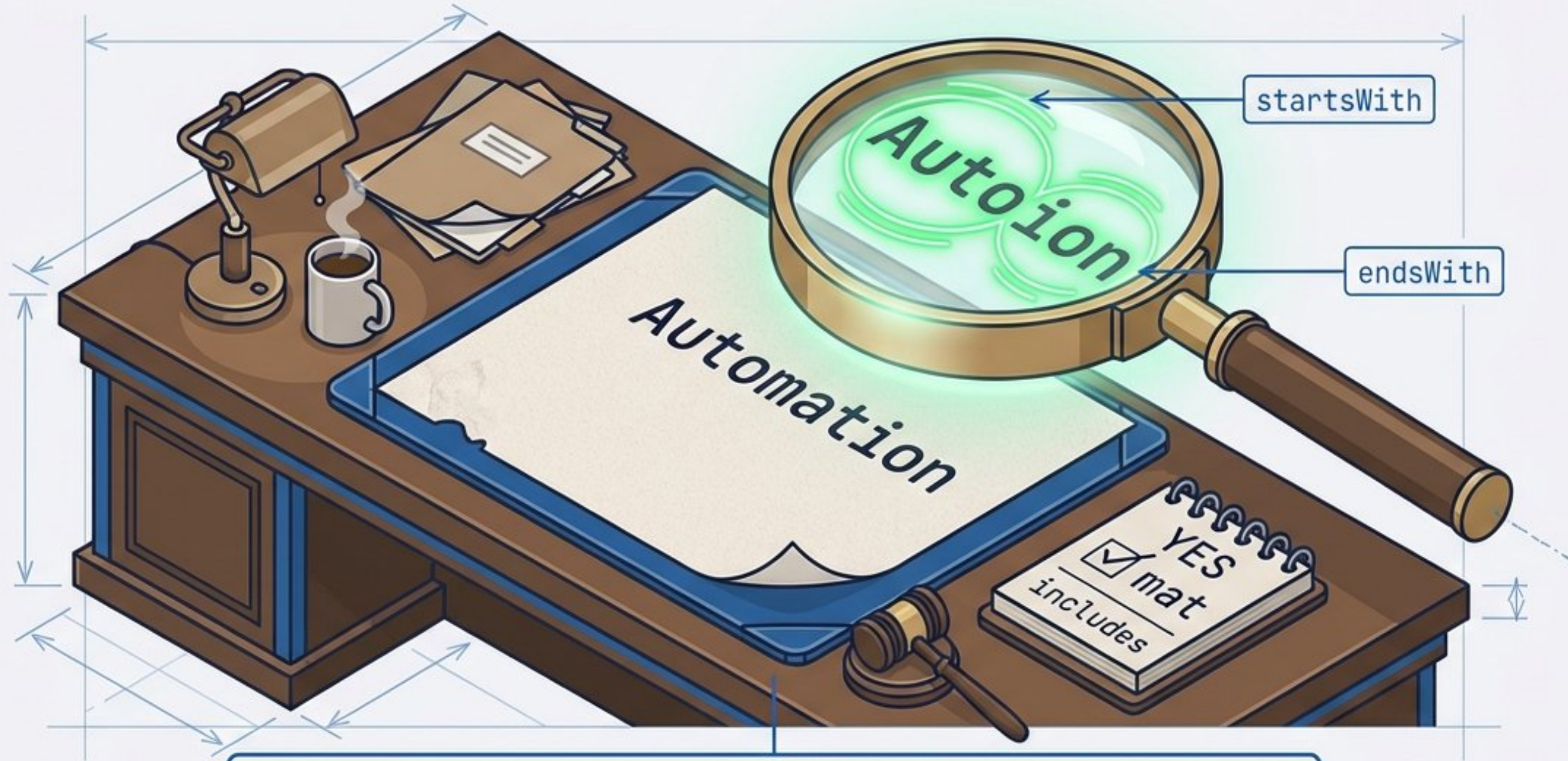


The Cleaner: Formatting Data

Input: Dirty String →
Output: Normalized Data



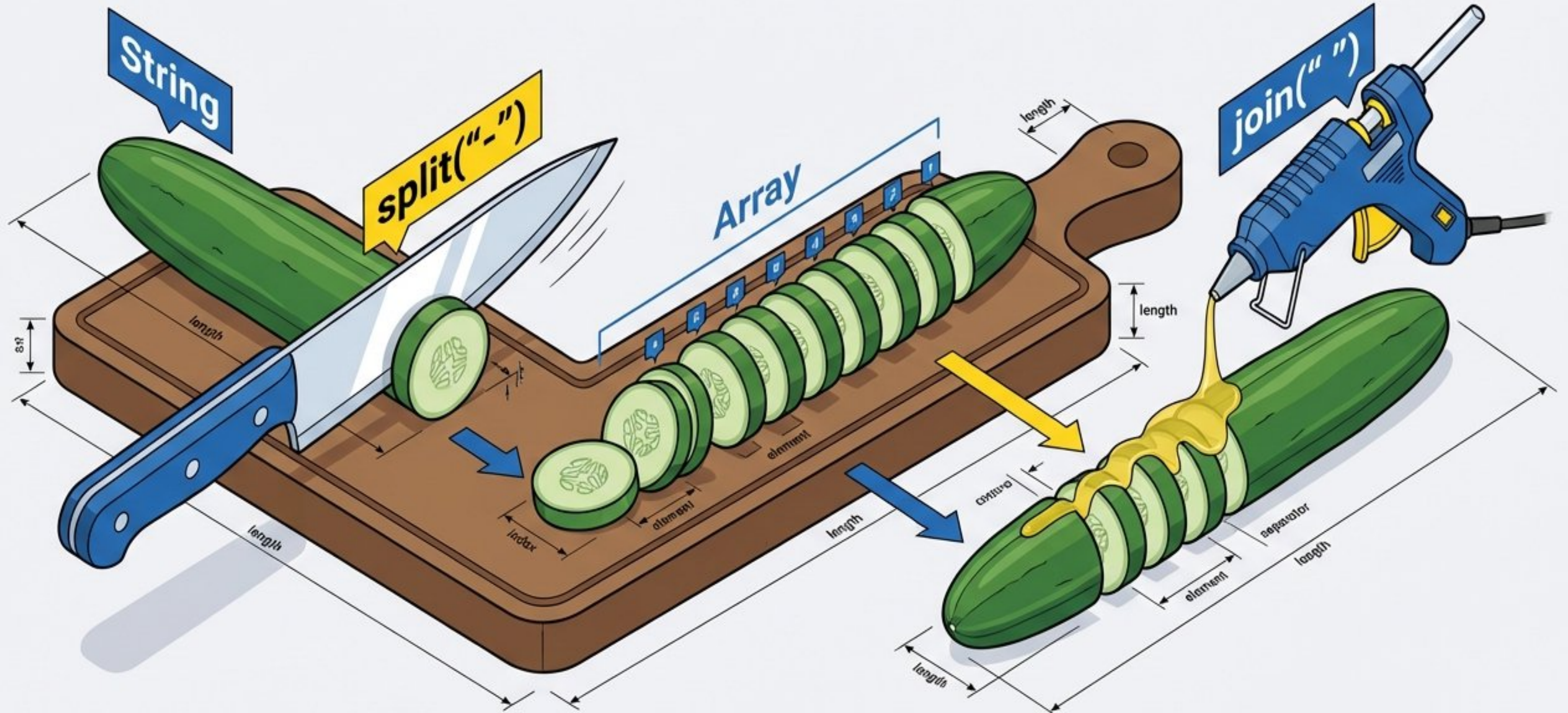
The Detective: Validation



⚠ JS uses `includes()`, NOT `contains()`.

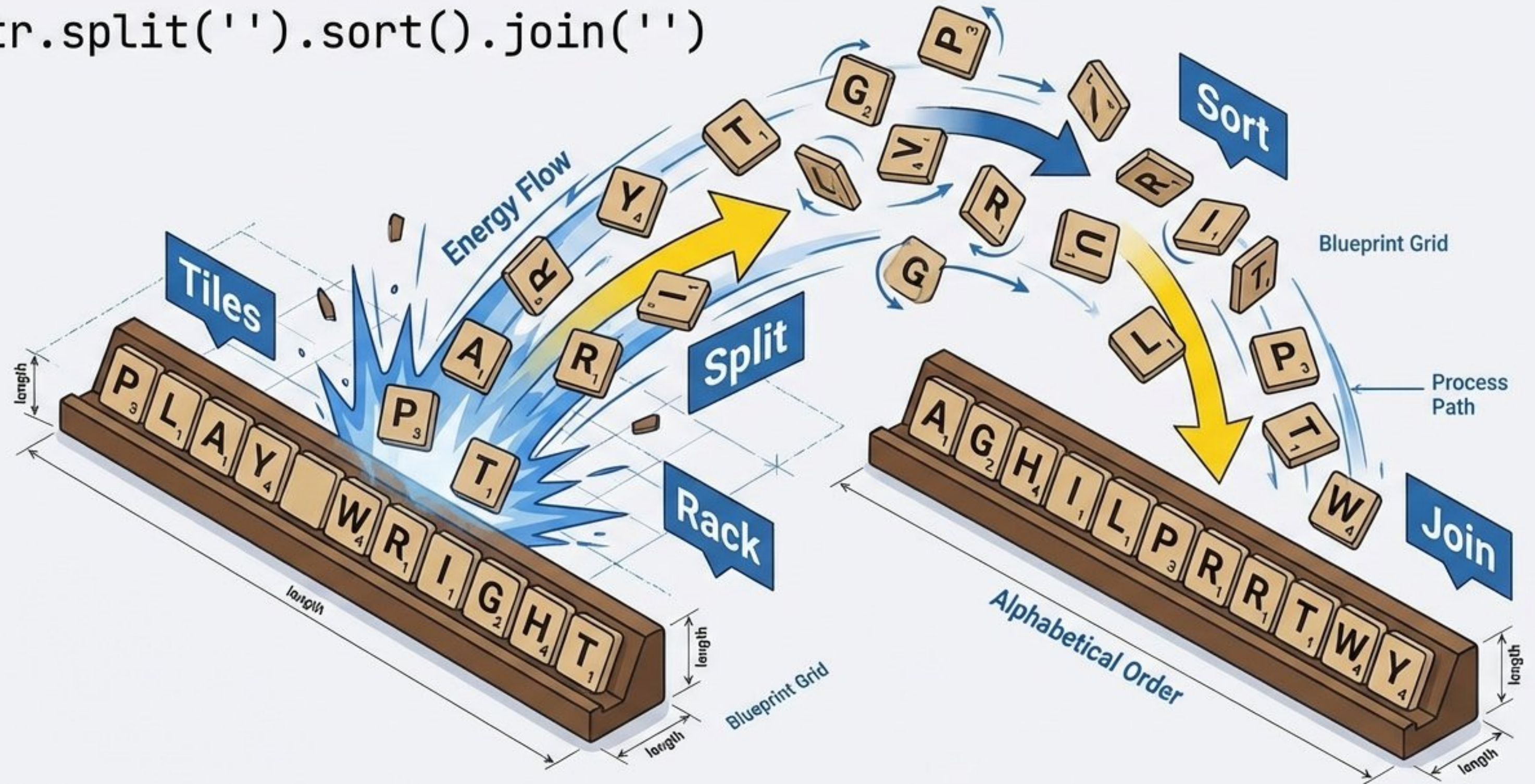
The Chopper: String $\leftrightarrow$ Array

`str.split("-")` vs `arr.join(" ")`



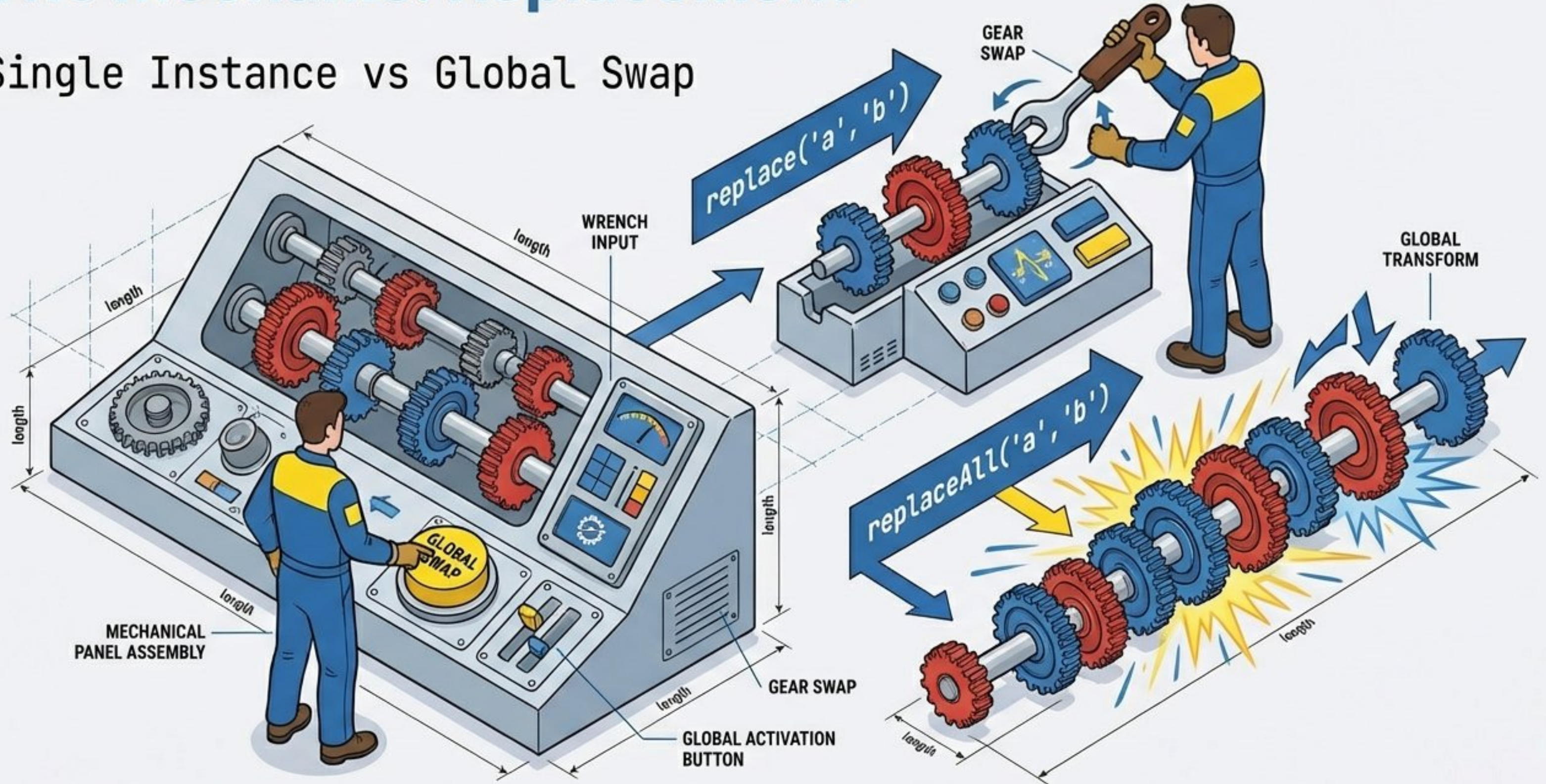
The Organizer: Sorting

```
str.split('').sort().join('')
```



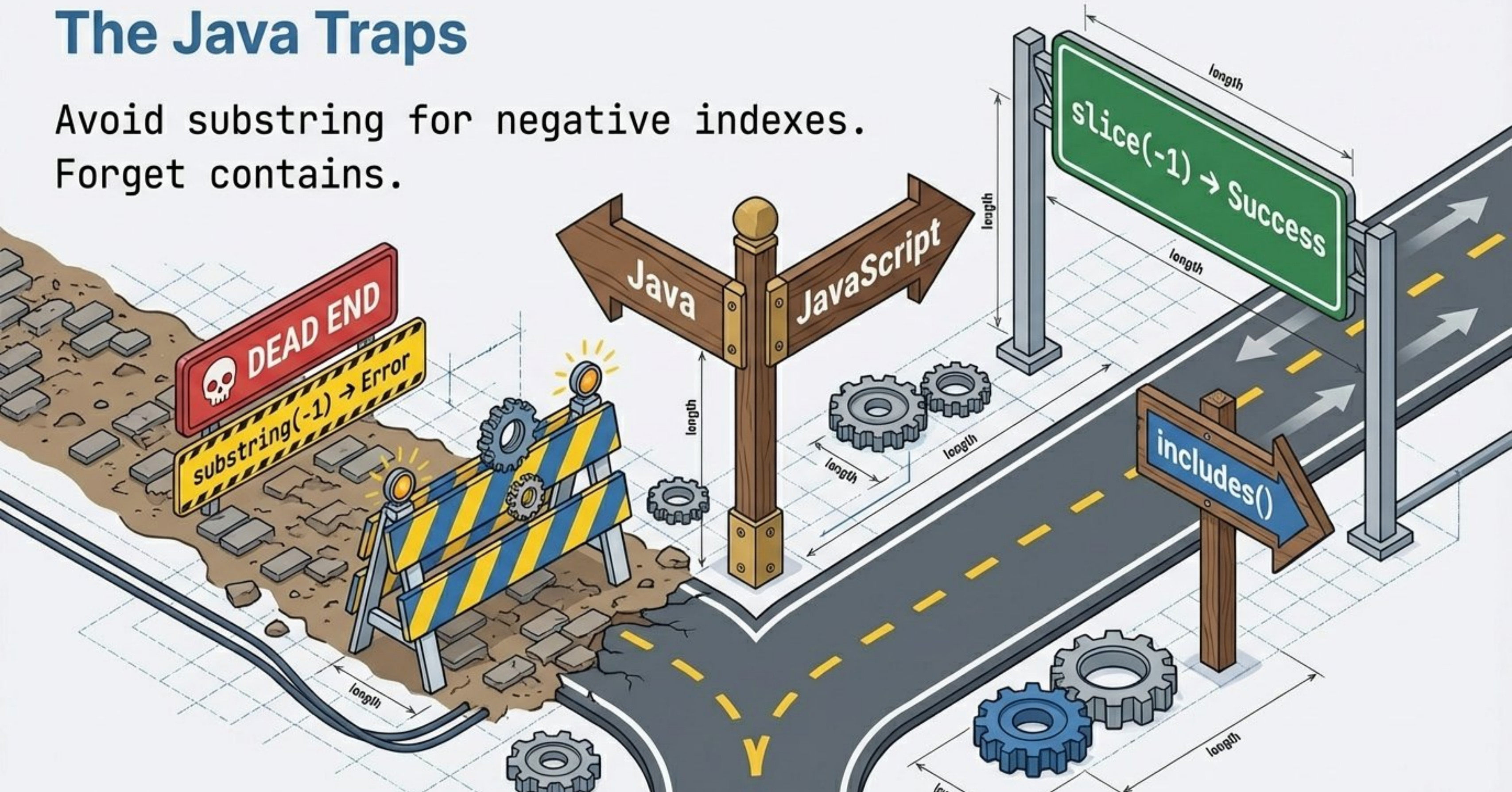
The Mechanic: Replacement

Single Instance vs Global Swap



The Java Traps

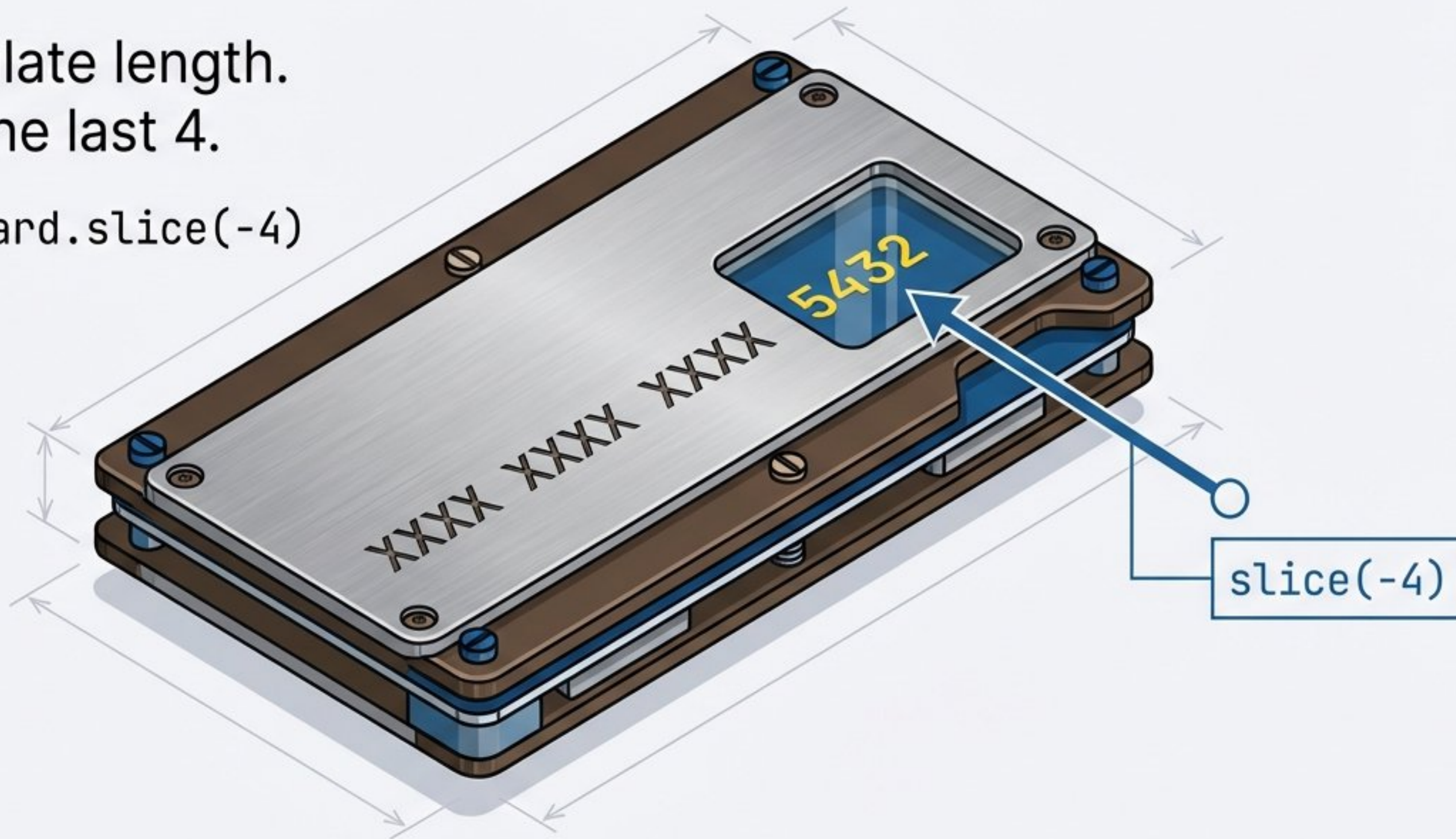
Avoid substring for negative indexes.
Forget contains.



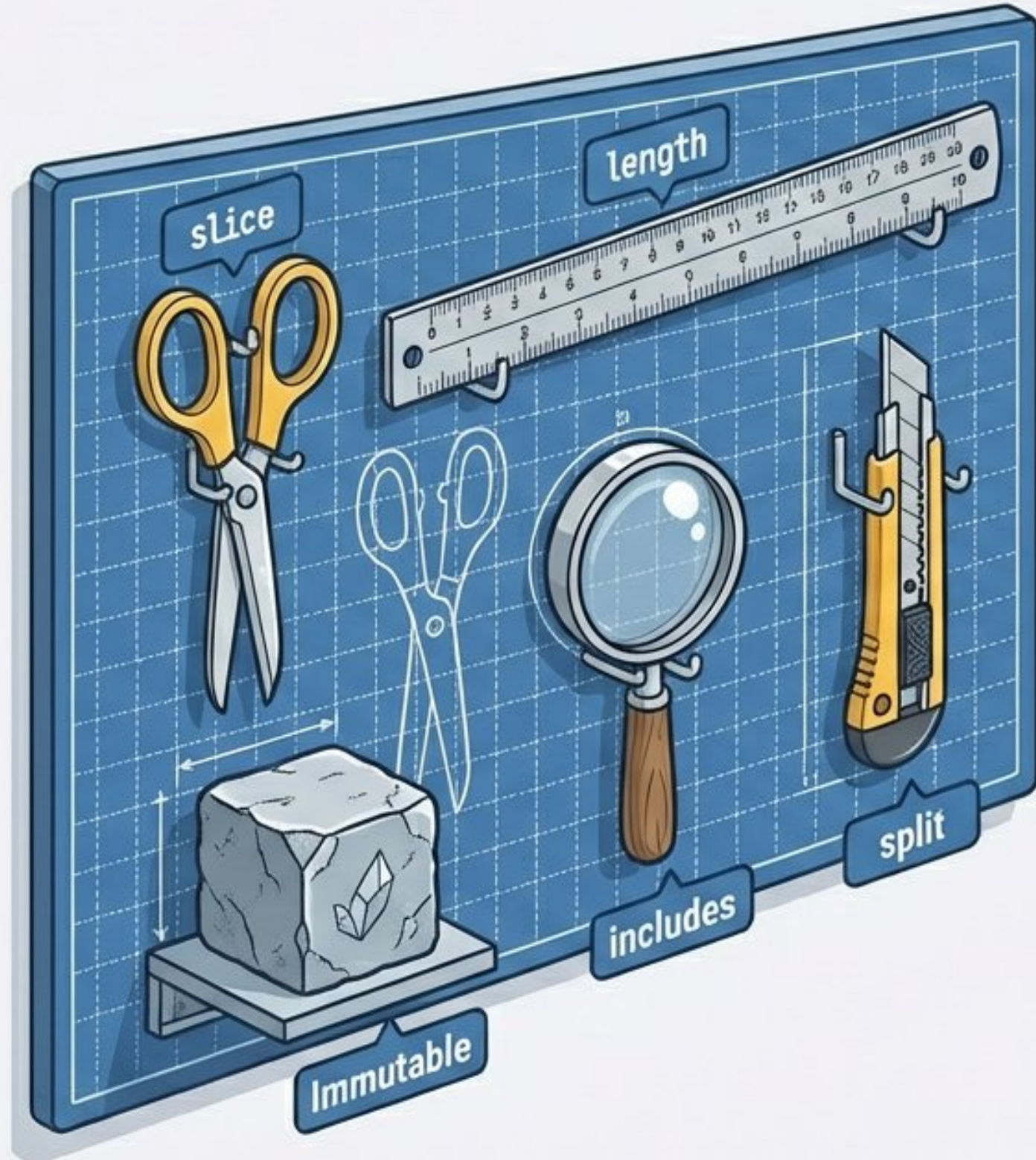
Use Case: Masking & Privacy

Don't calculate length.
Just take the last 4.

```
"XXXX " + card.slice(-4)
```



The Craftsman's Cheat Sheet



Reverse? ⚙️ Split/Sort/Join

Extract End? → Negative Slice

Check Content? → Includes